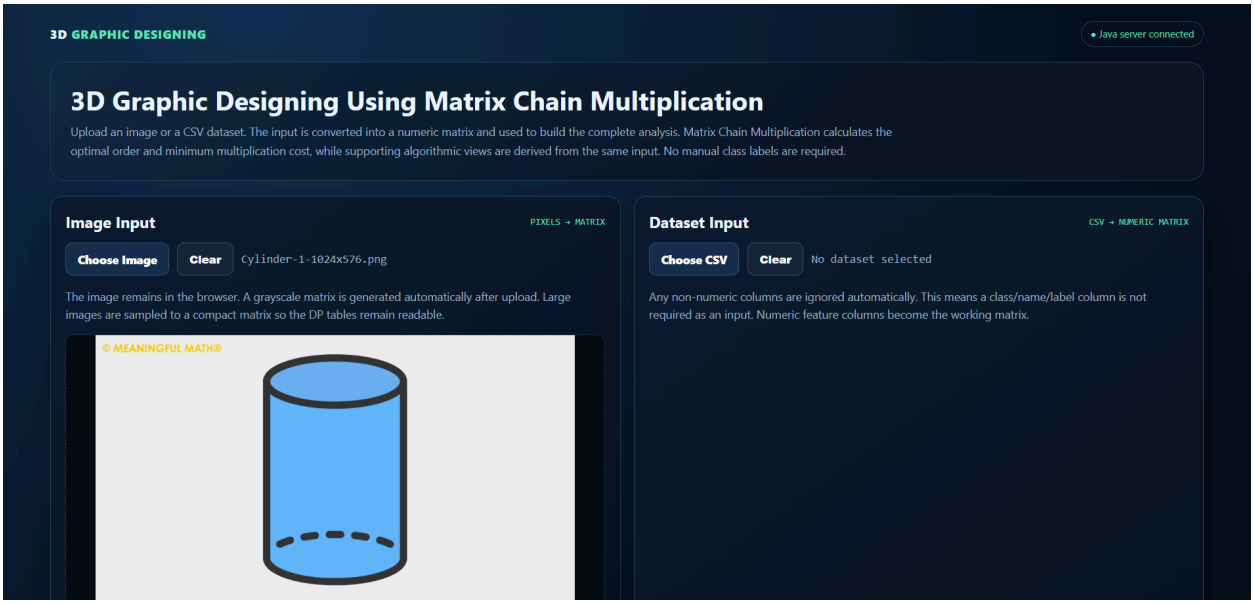
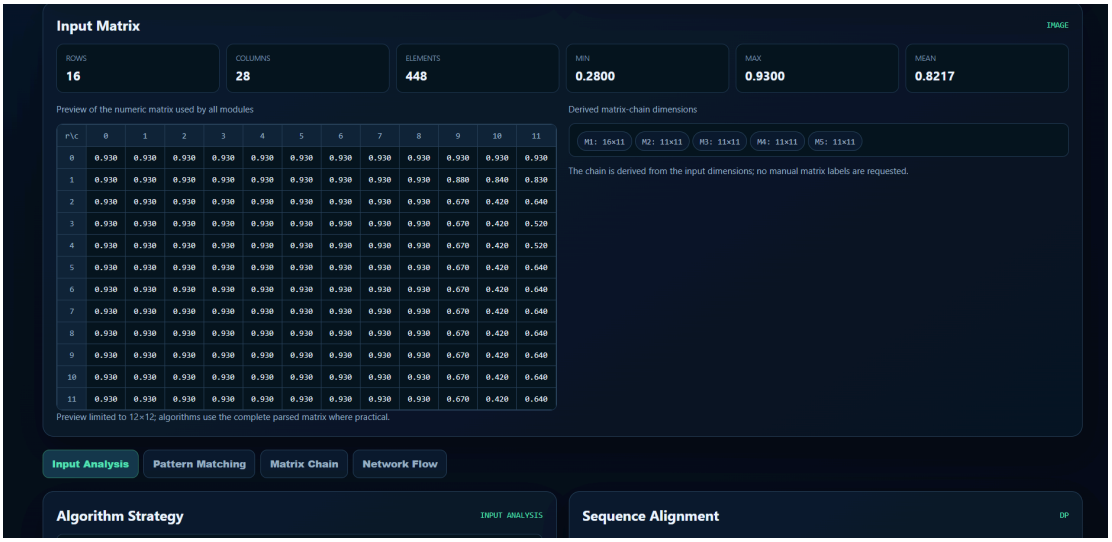
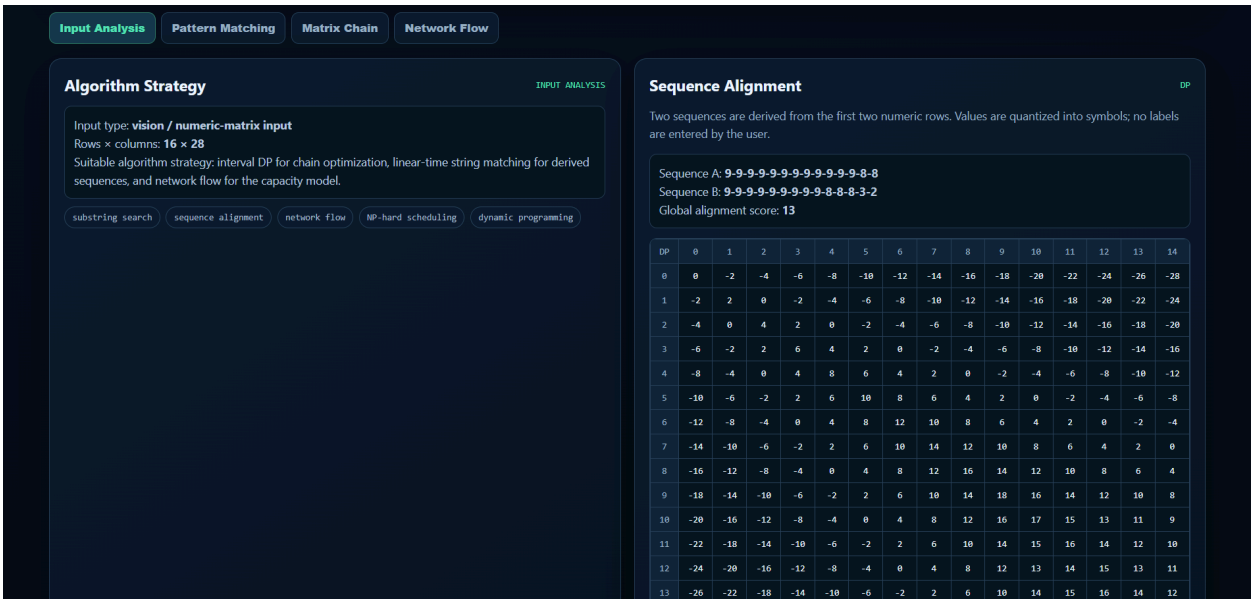
MATRIX CHAIN OPTIMIZER 3D
Interval Dynamic Programming Visualizer

Outputs:

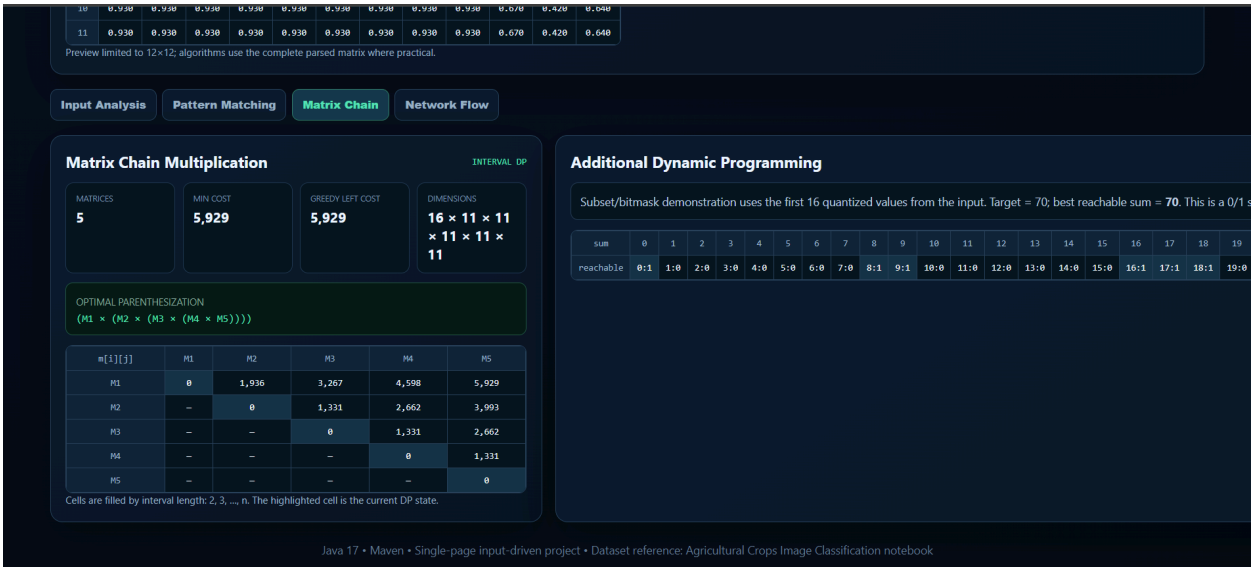




r\c	0	1	2	3	4	5	6	7	8	9	10	11
0	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930
1	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.880	0.840	0.830
2	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
3	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.520
4	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.520
5	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
6	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
7	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
8	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
9	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
10	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640
11	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640



DP	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20	-22	-24	-26	-28
1	-2	2	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20	-22	-24
2	-4	0	4	2	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20
3	-6	-2	2	6	4	2	0	-2	-4	-6	-8	-10	-12	-14	-16
4	-8	-4	0	4	8	6	4	2	0	-2	-4	-6	-8	-10	-12
5	-10	-6	-2	2	6	10	8	6	4	2	0	-2	-4	-6	-8
6	-12	-8	-4	0	4	8	12	10	8	6	4	2	0	-2	-4
7	-14	-10	-6	-2	2	6	10	14	12	10	8	6	4	2	0
8	-16	-12	-8	-4	0	4	8	12	16	14	12	10	8	6	4
9	-18	-14	-10	-6	-2	2	6	10	14	18	16	14	12	10	8
10	-20	-16	-12	-8	-4	0	4	8	12	16	17	15	13	11	9
11	-22	-18	-14	-10	-6	-2	2	6	10	14	15	16	14	12	10
12	-24	-20	-16	-12	-8	-4	0	4	8	12	13	14	15	13	11
13	-26	-22	-18	-14	-10	-6	-2	2	6	10	14	15	16	14	12



11	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.930	0.670	0.420	0.640

m[i][j]	M1	M2	M3	M4	M5
M1	0	1,936	3,267	4,598	5,929
M2	–	0	1,331	2,662	3,993
M3	–	–	0	1,331	2,662
M4	–	–	–	0	1,331
M5	–	–	–	–	0

sum	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
reachable	0:1	1:0	2:0	3:0	4:0	5:0	6:0	7:0	8:1	9:1	10:0	11:0	12:0	13:0	14:0	15:0	16:1	17:1	18:1	19:0

Input Analysis | Pattern Matching | Matrix Chain | **Network Flow**

Capacity Network

INPUT-DERIVED

The capacity network is derived directly from the numeric matrix. Source = 0; sink = 7; nodes = 8. Positive directed capacities are created from pairwise input values.

u\v	0	1	2	3	4	5	6	7
0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	1
2	0	0	0	1	1	1	1	1
3	0	0	0	0	1	1	1	1
4	0	0	0	0	0	1	1	1
5	0	0	0	0	0	0	1	1
6	0	0	0	0	0	0	0	1
7	0	0	0	0	0	0	0	0

Edmonds-Karp

MAX-FLOW / MIN-CUT

Maximum flow: **7**
Augmenting paths: **7**
Residual graph computed after all augmentations.

```
Step 1: 0 → 7  |  bottleneck 1  |  total 1
Step 2: 0 → 1 → 7  |  bottleneck 1  |  total 2
Step 3: 0 → 2 → 7  |  bottleneck 1  |  total 3
Step 4: 0 → 3 → 7  |  bottleneck 1  |  total 4
Step 5: 0 → 4 → 7  |  bottleneck 1  |  total 5
Step 6: 0 → 5 → 7  |  bottleneck 1  |  total 6
Step 7: 0 → 6 → 7  |  bottleneck 1  |  total 7
```